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Q7. (a) Discuss the Ageing and Senescence and describe the biological and social theories of Ageing 15

Ageing is defined as the process of getting old irrespective of when it occurs in the life span.

Senescence acc. to Strehler (1962) and Sir Peter Moaden is the decrease of survival capacity due to functional decline in the post reproductive phase.

Thus Senescence usually occurs after Ageing. Not all Ageing is senescence.

Ex: Childhood to Puberty is ageing but not senescence.

⇒ It is continuous, gradual and cumulative process.

⇒ Senescence is deleterious and ultimately results in death. Not the same for ageing.



## Biological Theories

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"Lifespan:

- ① Why we Age & why we don't have to  
- Prof. David Sinclair of Harvard University (2021)

→ Loss of epigenetic information

like Methylation & Histones leads to ageing. Without them cell doesn't know which genes to read.

Ex: Horvath Clock to tell biological age.

Keep body stressed to engage the protective mechanism to safeguard

the Epigenome. Ex: Heat, Exercise, etc.

### ② Programmable Theory

1) Death Gene: Activated at particular stage.

Ex: Monozygotic Twins have a similar life

Ex: Hunter of Pak live long life.

### 2) Endocrine Theory

3) Immunogenetic: By Wallace & Burnet  
Theories

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### ③ Wean & Thean Theory

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- 1) Telomere Shortening
- 2) Cellular & Tissue : August Wicman (1955)
- 3) Freed Radical (Damage: Denham (1950) Bauman)  
→ But not all are bad; in fact may help in cellular comm<sup>n</sup> & destroy bacteria  
→ Cross-Linked protein harmful acc. to Bjorkstein (1942)

### <Social Theory>

- ① Cummins & Henry in "Getting old 1961" gave Disengagement Theory.

As we get old → loose connection.

Continuum: People not doing it so not universal theory now.

- 81/✓
- ② Continuity Theory: Robert Atchley  
People try to maintain old life.  
Ex: use spectacles to continue reading books

Continuum: Does not take into account the Chronic illnesses

- ③ Activity Theory (1960's) by Robert J. Havighill  
Opposite of first<sup>st</sup> Theory.

while Thus Biological theories are conflict, while social theories not universal → Open question